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Paleomagnetism of some Indian kimberlites and lamproites

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Abstract

Consistent vectors have been isolated by AF demagnetization of 22 cores from two sites in Proterozoic kimberlites in the Wajrakarur district of the Dharwar Craton ($N=2$, $D=156$, $I=-71$, $K=1596$, $\alpha=6.3$, Pole WAJ 45.4°S, 121.5°W),and 12 cores from two sites from the mine at Majhgawan in the Panna district in the Aravalli craton ($N=2$, $D=34$, $I=-27$, $k=53$, $\alpha=35.4$, Pole MAJ, 38.9°N, 216.5°E). Surprisingly, these 1200–1000 Ma poles fall close to roughly coeval points on the APW curve for the Kalahari Craton in a Smith–Hallam Gondwana reconstruction. If the poles and their ages are correct, these data constrain Mesoproterozoic supercontinent reconstructions.

1. Introduction

In 1980/81 oriented samples for paleomagnetic study were drilled from seven kimberlite or lamproite occurrences in south and central India (see Fig. 1). Previously published radiometric data from several of these occurrences (Table 1) suggest that they were all intruded in late Proterozoic time, and hence were roughly coeval with the Premier swarm in South Africa, for which paleomagnetic data had already been published (Jones, 1968; Hargraves and Onstott, 1980; see Doppelhammer and Hargraves, 1994). If good paleomagnetic data could be obtained from the Indian occurrences, and the ages could be refined, then the paleogeographic contiguity, or otherwise, of India with Africa at that time could be ascertained. John Bristow reports (pers. commun., 1985) an age of 1079 Ma obtained by U–Pb measurements on perovskite separated from a Muligiripalle sample collected in this study.

2. Diamond occurrences in India

Diamonds were known to occur in sandstones and conglomerates as well as in the modern river gravels of India perhaps as early as 800 B.C. India was trading in diamonds at least as early as the 4th century B.C., but little is known about this early period of diamond mining as nothing was written about it until the jeweller Jean-Baptiste Tavernier visited the mines in the mid-17th century. Many of the workings he examined were located along the Krishna River nearly 150 km southeast of Golconda, (near present-day Hyderabad), the headquarters for the diamond merchants (Bruton, 1978). The famous Koh-i-Nur, Great Mogul and Tavernier blue diamonds are believed to have come from the Kollur mine, situated in this region (Smith, 1972; Bruton, 1978). In northern India, near Panna, additional diamond diggings exist that were apparently unknown to Tavernier, but which are

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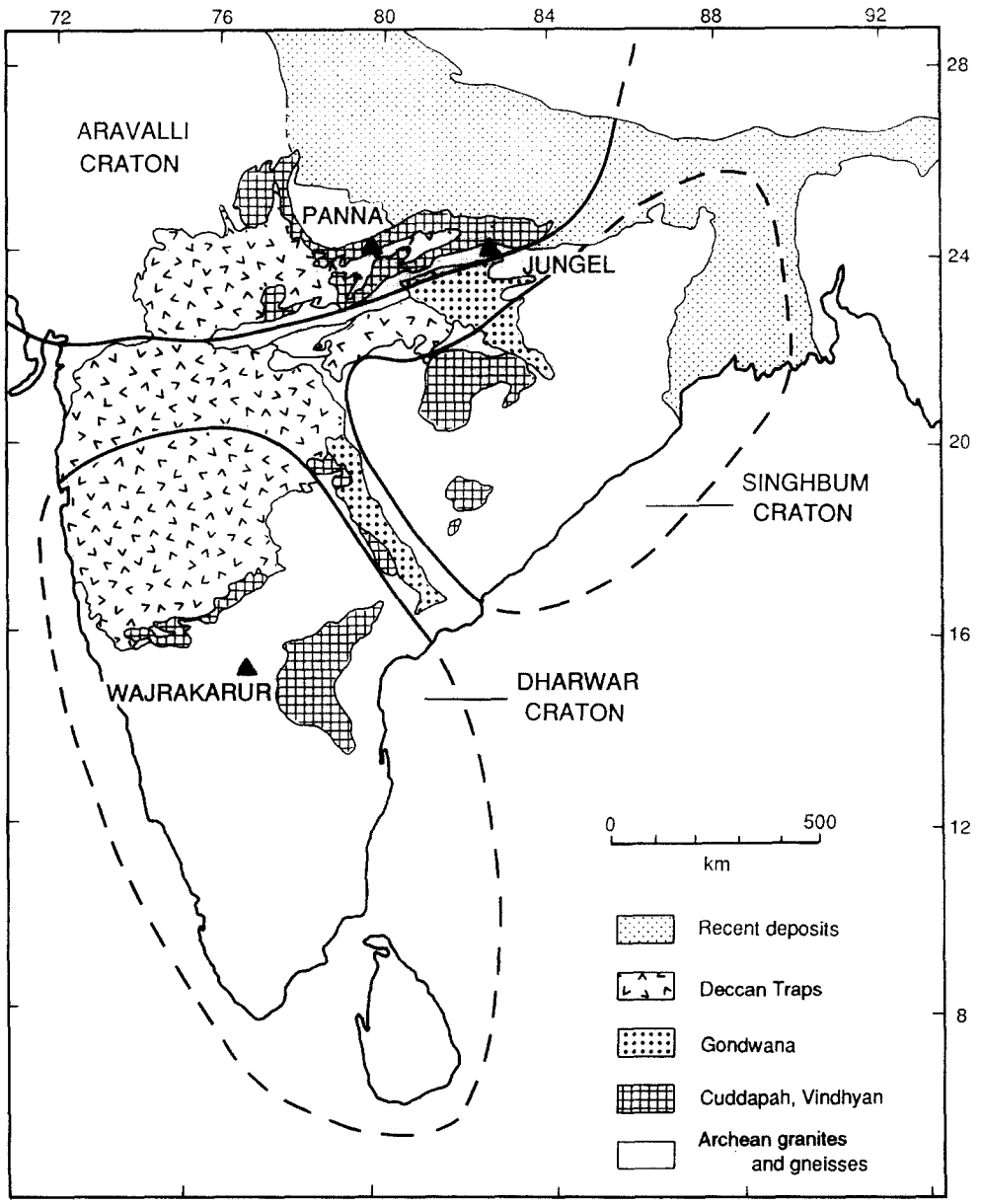


Fig. 1. Generalized geologic map of India showing location of kimberlite/lamproite districts relative to the hypothesized separate cratons (Naqvi, 1974; Sarkar, 1980).

known to have been worked during the 16th century if not earlier (Sinor, 1930). At Panna, the diamonds are found in sandstones and conglomerates, in alluvium and rain wash derived from the conglomerates and in a volcanic plug that has now been classified as lamproite (Scott Smith, 1989) and was recently being mined. In addition, the Angore pipe and the Hinota pipe are now known to exist in this region. A similar kimberlite pipe was discovered in southern India near Wajrakarur in 1880 (Chatterjee, 1964). More pipes have been uncovered in this area recently

Table 1
Kimerlite age data

Site	Method	Age (m.v.)	Reference
Indian pipes			
Wajrakarur	Rb–Sr whole rock	1350±294	Paul, 1979.
	Rb–Sr whole rock, mica	1116	Crawford and Compston, 1973.
	K–Ar whole rock	840±30 to 1023±40	Paul et al., 1975.
Lattavaram	K–Ar whole rock	933	Paul et al., 1975.
	Rb–Sr whole rock	1505	Paul, 1979.
Muligiripalle	U–Pb on perovskite	1079	John Bristow, pers. commun. 1985.
Chelima Diatreme	Rb–Sr whole rock	1199	Crawford and Compston, 1973.
Cuddapah dikes	Rb–Sr whole rock	959±110	Crawford and Compston, 1973.
Majhgawan	Rb–Sr whole rock	1630±353	Paul, 1979.
	Rb–Sr mica	1116±12	Crawford and Compston, 1970.
	K–Ar mica	1056	Crawford and Compston, 1970.
	K–Ar whole rock	947±30 to 1170±46	Paul et al., 1975.
Jungel	K–Ar whole rock	919±21	Balasubramanyam et al., 1978.

Note: the constant $\lambda^{87}Rb = 1.42 \times 10^{-11}\ yr^{-1}$ is used for all Rb–Sr dates.

(Scott Smith, 1989). Fig. 1 is a map showing the main kimberlite or lamproite occurrences in India.

3. Field and laboratory methods

One inch (2.5 cm) diameter cores were collected from seven "kimberlitic" bodies (Table 2) by means of a gasoline-powered drill, and oriented with a sun and magnetic compass. The remanent magnetism (RM) of at least one specimen from each core was measured on a Schonstedt DSM spinner magnetometer. Alternating field (AF) demagnetization was performed on a custom-built apparatus with maximum field of 120 mT. Thermal demagnetization of selected samples was performed in a Schonstedt TSD-1. All this equipment is housed in a magnetically shielded room. The results were analyzed by Zijderveld diagrams and principal component analysis (Kirschvink, 1980). Polished thin sections were made of representative samples from each site.

Table 2
Paleomagnetic results from Indian kimberlitic bodies

Site	Location	Cores	Regression interval	n	D	I	k	α_{95}
2	Wajrakarur	9	unstable					
3	Wajrakarur	9	unstable					
5	Lattavaram	11	SEP[a], 30–70 mT	10	160.5	−70.6	7	20.6
6	Muligiripalle	11	SEP, 30–70 mT	9	152.1	−71.5	8	21.4
24	Majhgawan	6	~7 steps, N–30 mT	6	40.2	−32.3	28	12.8
25	Majhgawan	6	~8 steps, N–40 mT	6	27.4	−21.6	54	9.2
26	Angore	8	SEP	6	52.6	26.4	27	13.1
27	Jungel	4	SEP	3	354.9	32.6	187	9.1

[a]SEP = stable end point.

4. Petrology

Since our initial petrographic study, many of our samples have been studied by experts in the petrology and petrochemistry of kimberlites and related rocks. What follows is an abbreviated summary of the collective findings (Bergman, 1987; Scott Smith, 1989).

The four pipes sampled near Wajrakarur run the gamut from fresh to completely altered. Sites 5 (Pipe 4, Lattavaram) and 6 (Pipe 5, Muligiripalle) are relatively fresh; site 2 (Pipe 2, Wajrakarur) is partially and site 3 (Pipe 1, Wajrakarur) is extremely altered. Despite local variation, Scott Smith (1989) describes these as mica-bearing monticellite kimberlite. Pipe 5 (Lattavaram) has fresh olivine, only partially serpentinized. Pipe 6 (Muligiripalle) has above average phlogopite, olivine is partially serpentinized, and calcite is present.

The Majhgawan pipe in the Panna district (Sites 25 and 26) (Mathur and Singh, 1971; Akella et al., 1979) which has been mined (Fig. 1), has been classified as a lamproite by Scott Smith (1989), who describes the material as lapilli tuff, "composed of olivine macrocrysts and phenocrysts set in glassy groundmasses containing mica, vesicles, apatite and altered perovskite". The olivine is almost entirely serpentinized and surrounded by fine grains of perovskite. Fine magnetite grains are disseminated throughout the matrix, as well as in expanded mica flakes, and is variably altered to hematite.

The Angore (Nane, 1971) and Jungel (Murthy et al., 1980) bodies are extensively carbonated and altered, and their classification as kimberlites or lamproites has been questioned by Scott Smith (1989).

5. Magnetic results

The final results are summarized in Table 2. Overall, the samples from southern India were found to be much less stable than those from the north. Samples from the Wajrakarur Pipes 1 and 2 (Sites 3 and 2, respectively) which were extremely altered, were weak with widely scattered NRM's ($k<3$). Upon demagnetization, the intensity of the vector diminished rapidly (MDF <5 mT) and at each site they appeared to separate into two distinct groups which clustered best after 15 mT AF. Thereafter the vector became even more scattered. At both sites, one of the two clusters closely approximated the present earth's field, and is presumably related to a modern overprint. The second clusters at the two sites (indistinguishable within the large error) are southerly with positive inclination, but the significance of these vectors is questionable.

The NRM's at Site 5 (Lattavaram, Pipe 4) and Site 6 (Muligiripalle, Pipe 5) were also hopelessly scattered, but the cores were somewhat more stable than those from Wajarakrur (M.D.F.'s of ~ 7.5 mT). Upon AF demagnetization the cores from both of these sites converged above ~ 30 mT toward a stable end point vector which was southerly with steep negative inclination (see Table 2 and Fig. 2). In these cores, the acquisition of rotational remanent magnetism (RRM, Stephenson, 1980a,b, 1983) components became noticeable above 30 mT. Repeat measurements inserting the cores in the AF demagnetizer with alternating orientation showed that the spurious component torqued the residual vector first one way then another (see Fig. 2). Averaging of at least 6 of these oscillating vector measurements for each core gave the results reported in Table 2. Clearly, these vectors were rather scattered, resulting in site k's of 7 and 8, respectively; the means from each site, however, were very similar.

The samples from the Majhgawan mine (Sites 24 and 25) gave consistent NRM's which remained relatively constant to 30 and 40 mT, respectively (Fig. 2). On average, seven demagnetization steps were regressed yielding vectors, with MAD <15. The vectors for two of the six samples at site 25 were reversed; the reason for this was not established. The average vectors from the two sites were very similar.

The NRM of the intensely carbonatized samples from Angore was extremely stable, and stable-end point means were calculated. The vectors of two of the samples, however, were

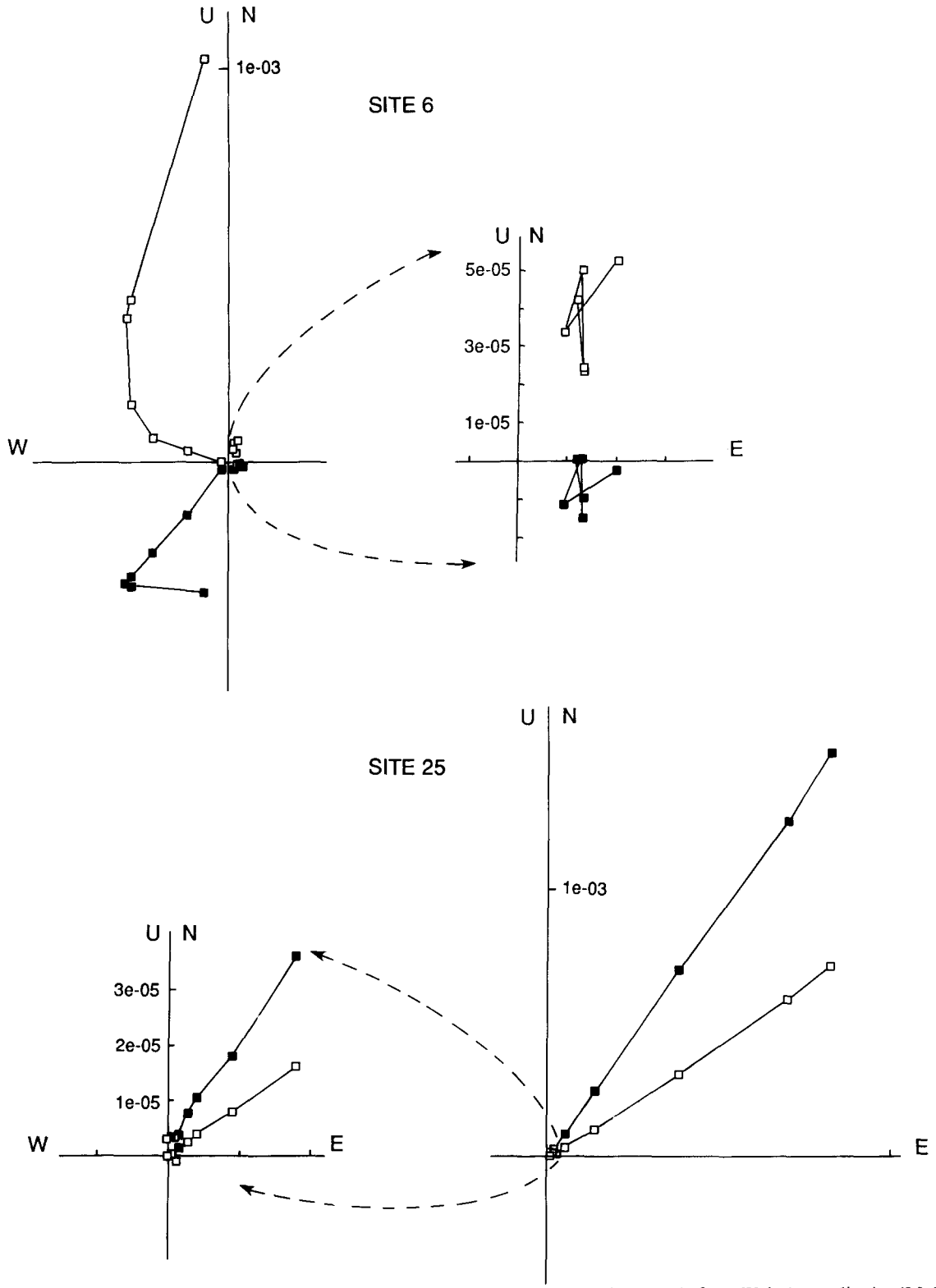


Fig. 2. Zijderveld diagrams illustrating AF demagnetion behavior of a representative sample from Wajrakarur district (Muligiripalle, Site 6) and Panna district (Majhgawan, Site 25). Note "stable" end point reached in samples from Wajrakarur and linear regressing vector from Majhgawan.

distinctly different from the remaining five. The value of this result is not clear.

The three completely serpentinized samples from a shaft on a pipe at Jungel yielded a stable and consistent vector, which was, however, very close to the present-day field. It too will not be considered further.

6. Comparison with previous paleomagnetic studies

Paleomagnetic studies of kimberlites in the Wajrakarur district have been reported by Anand (1971) and Poornachandra Rao et al. (1984). Anand's study of Pipes 1 and 2 at Wajrakarur and Pipes 3 and 4 at Lattavaram (note that Pipe 3 was not sampled in the present study) resulted in mean vector after 12 and 8 mT AF demagnetization, respectively, (and considerable data selection) of 104°, −39° $\alpha_{95}=11°$, and 158°, +42°, $\alpha_{95}=40°$. These results are quite unlike those derived in this study and are considered to be superseded.

Poornachandra Rao et al. (1984) report results from the same pipes sampled in our study. (Pipes 1, 2, Wajrakarur; Pipe 4, Lattavaram; Pipe 5, Muligiripalle). Their NRM results were entirely similar to ours (highly scattered) and they also found the remanence to be highly unstable. As their pilot study AF demagnetization (up to 80 mT) of 12 specimens from the four pipes showed no evidence of grouping, they demagnetized the remainder of their collection to fields no higher than 15 mT (op.cit., p. 110). Their samples from Lattavaram (Pipe 4, our Site 5) apparently were particularly susceptible to RRM acquisition, and they obtained no useful data from these cores. In their analysis of the data for the other three sites, they determined mean vectors using the "minimum scatter" method, and report results which, while superficially similar to ours (which we considered unacceptable) for Pipes 1 and 2 (Wajrakarur), were totally unlike ours for Pipe 5 (Muligiripalle; $N=7$ samples, $n=23$ cores; $D=320°$, $I=+40$, $k=29.9$, versus $N=9$, $D=152.1$, $I=-71.5$, $k=8$). We consider our demagnetization treatment (to 70 mT) and analytical procedure to be more robust than those of Poornachandra Rao et al. (1984). Furthermore, the fact that we obtained very similar site-mean vectors after demagnetization in high fields from both Lattavaram and Muligiripalle (despite considerable within site scatter, see Table 2), inspires confidence that these are "characteristic" remanent vectors.

Table 3
Mean paleomagnetic poles

		Poles		k	α_{95}
		lat.	long		
Lattavaram, Muligiripalle (Wajrakarur district) WAJ	$N=2$	45.45°S	238.5°E	537	10.8
Majhgawan (Panna district) MAJ	$N=2$	38.9°N	216.5°E	68	30.9
Rotated to Africa in Smith–Hallam (1970) reconstruction, and polarity reversed.					
Compare:					
Wajrakarur, WAJ		26.0°N	65.7°E		
Majhgawan, MAJ		10.3°N	344.6°E		
Premier, PMR		51.0°N	37.0°E		
National, NAT		21.0°N	65.0°E		
Schuller, SCH		33.4°N	53.5°E		
Franspoort, FRA		49.1°N	48.8°E		
Premier average, PAV		30.0°N	52.5°E[a]		

[a]Doppelhammer and Hargraves, 1994.

7. Discussion

The similar results obtained from Sites 5 (Lattavaram) and 6 (Muligiripalle) in the Wajrakarur district, and Sites 24 and 25 from the Majhgawan mine in the Panna district encourages the conclusion that these vectors may have paleogeographic significance. They have been averaged ($N=2$ at each locality), and the poles calculated are listed in Table 3. With only two sites, however, their validity as paleomagnetic poles can be questioned. But clearly, the vectors/poles from these two localities are ~90° apart, and thus there is no paleomagnetic evidence to suggest that these kimberlitic bodies were intruded penecontemporaneously. Even if the Aravalli and Dharwar cratons (Fig. 1) had then been in their present position with respect to one another, this separation suggests a significant difference in their time of emplacement (or magnetization).

In light of the contemporary trend toward inferring the repeated assembly and break-up of supercontinents (e.g. Dalziel, 1991; Hoffman, 1991), there is no knowing where India was located with respect to any of the cratons in Africa ~1 Ga ago. Nevertheless, given the rough penecontemporaneity of the emplacement of these kimberlitic bodies in India with the Premier swarm in Southern Africa, one may wonder

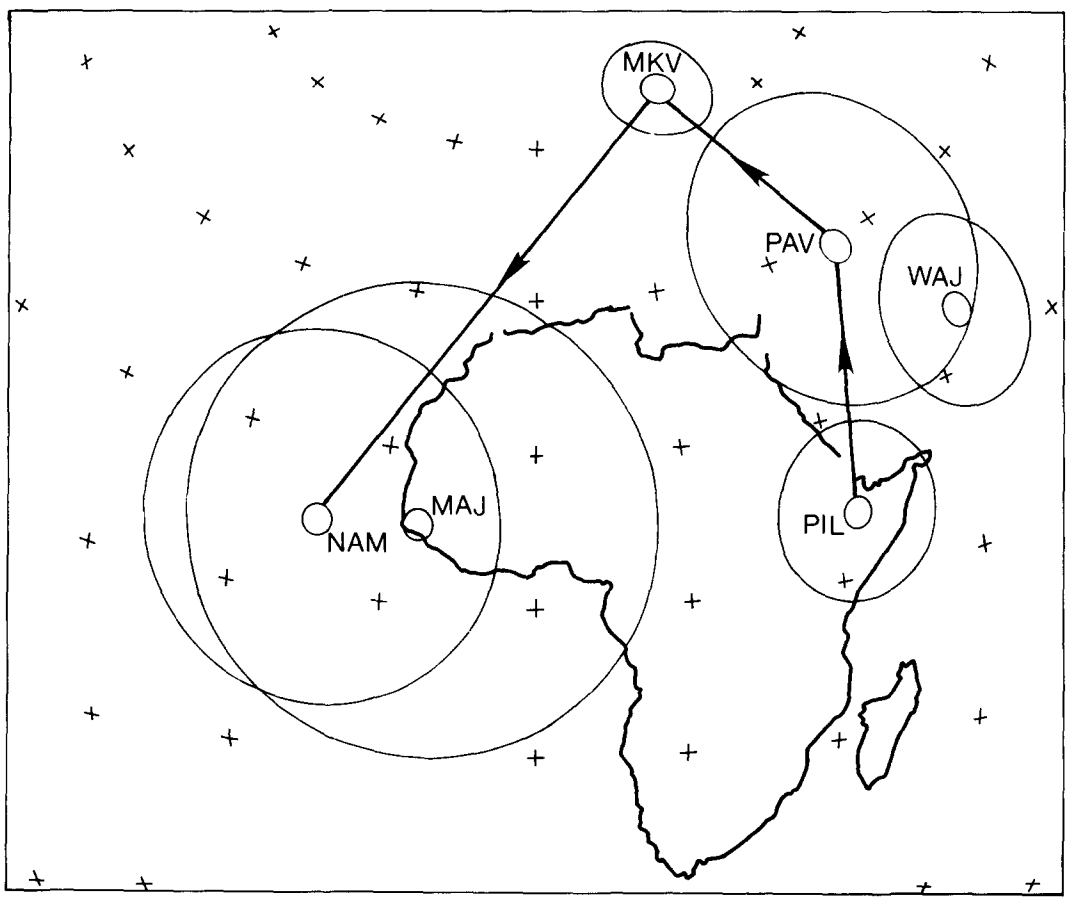


Fig. 3. Neoproterozoic (1300–900 Ma) APW curve for the Kalahari craton (Pilansberg—*PIL*, ~1300Ma, Gough, 1956; Van Niekerk, 1962; Premier average—*PAV*, ~1180 Ma, Doppelhammer and Hargraves, 1994; Umkondo Average—*MKV*, ~1080 Ma, Allsopp et al., 1989; Namaqualand Average—*NAM*, ~1000 Ma, Renne et al., 1990) and the two Indian poles (Wajrakarur—*WAJ*, Majhgawan–*MAJ*) after restoration to Africa in a Smith–Hallam (1970) reconstruction.

where these poles would fall in a Gondwana reconstruction relative to the Kalahari Craton APW curve for that time frame. In a Smith–Hallam (1970) reconstruction (Table 3), illustrated in Fig. 3, it can be seen that the Wajrakarur pole falls remarkably close to the Premier pole average and the Majhgawan pole falls close to the 1000 Ma end of the Kalahari APW path. Whether this has paleogeographic significance, or results from inadequate paleomagnetic sampling and is a mere coincidence, must await more detailed determination of APW for India.

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